Network Position and Academic Performance

A Centrality-Augmented Replication of Smirnov & Thurner (2017)

ECC3841 Network Economics — Research Brief

Intro

Academic performance is shaped not only by individual effort and ability but by the social environment a student is embedded in. A growing empirical literature shows that friendship networks matter for outcomes like GPA — but *how* they matter is still contested. One well-known account, Smirnov & Thurner (2017), used a large, real, time-evolving "who-likes-whom" network of Russian high-school and university students to argue that friendship's effect on GPA similarity is driven almost entirely by **selection**: students change who they are friends with to keep their friend group's GPA close to their own, rather than being pulled up or down by their friends' performance. Their evidence for this is compelling, but their method treats the network only as a device for finding out *who a student's friends are* and averaging those friends' GPA. It never asks where a student **sits** within the larger shape of the network — whether they are a hub that many people independently like, or a node reachable only through long, indirect chains through many different social pockets. That second question — whether **network position**, not just **network composition**, independently shapes academic performance — is what this project investigates, using the exact same publicly available replication data Smirnov & Thurner released.

Research Question

> *Net of the friend-selection mechanism that Smirnov & Thurner (2017) identify, does a student's* ***position*** *in the friendship network — specifically, the distinction between being* directly popular *(liked by many people) and having* broad indirect reach *(connected, through longer chains, to many different and loosely-linked parts of the network) — independently predict GPA? And does this effect hold uniformly across a more socially stable environment (high school) and a more socially fluid one (university)?*

Main Idea

The "traditional" (Smirnov & Thurner) idea:

Friend selection $\rightarrow$ friend-group GPA similarity $\rightarrow$ observed homophily in GPA

Their method only ever asks: *who is this student connected to, and what is the average GPA of those specific people?* Two students with identical direct friends would receive an identical score under their method, no matter how those friends' own social worlds differ.

What we aim to analyse:

Network position (direct popularity vs. indirect reach) $\rightarrow$
academic performance, net of friend-group composition

For instance, suppose two students have an identical direct friend group of similar size and similar average GPA (so Smirnov & Thurner's method would treat them as equivalent):

- **Student 1** is liked by many classmates directly — their popularity is concentrated in one fairly tight-knit circle.
- **Student 2** is liked by fewer people directly, but is connected — through friends of friends of friends — to many different, loosely-linked social pockets across the school. Their *reach* is wide, even though their *direct* popularity is unremarkable.

Should we expect Student 1 and Student 2 to have the same GPA outcome, simply because their immediate friend group's average GPA looks the same? Our answer, developed below, is **no** — and the direction of the difference is not what a naive "more connections is better" intuition would predict.

Predictions

1. Replicating Smirnov & Thurner: (a) a friend's *past* GPA will not predict a student's *future* GPA once the student's own past GPA is controlled for (ruling out direct peer contagion); (b) newly-formed friendships will show a smaller GPA gap than friendships that are dropped (evidence of selection).
2. Katz–Bonacich centrality — a network-position measure entirely absent from the original paper — will retain independent explanatory power for GPA after controlling for the friend-composition channel Smirnov & Thurner identify.
3. The selection effect (prediction 1b) will be stronger in the university sample than in the high-school sample, reflecting university students' greater freedom to reshape their social circle (new courses, clubs, dorms).
4. The independent network-position effect (prediction 2) will be stronger in the high-school sample than the university sample, reflecting a more stable, longer-accumulated social structure.

Model Structure

Nodes: individual students (655 high-school students; 1,200–1,549 students per university cohort — freshmen, sophomores, juniors, seniors).

Edges: directed "like" ties reconstructed from a social-networking site — an edge from i to j exists if i gave j at least one "like" within a given 3-month window. Networks are observed at 2–14 snapshots per cohort (38 snapshots in total) and are **not symmetric**: only 24–27% of ties are mutually reciprocated, so a "friend" in this data is really "someone I chose to like," not a confirmed mutual relationship.

Node attributes:

1. GPA (outcome variable; time-varying for the high-school cohort, a single measurement for the four university cohorts)

2. Network position: in-degree (direct popularity), out-degree, Katz–Bonacich centrality (computed at multiple attenuation levels), betweenness, eigenvector centrality
3. Cohort / level: high_school vs. university (and the specific class-year)

Related Studies

1. AddHealth-based peer effects (IZA DP 3859). *Nodes:* individual students. *Edges:* real friendship nominations from AddHealth — each student named their best friends from a school roster. *Node attribute:* school performance / GPA, the main outcome variable.

2. Family background and adult mental health (IZA DP 13680). Studies the influence of family background on mental health, but constructs no individual-level network. *Nodes:* individual students. *Edges:* shared membership in the same school–cohort–gender reference group (a complete network within each group), not individual friendship ties. *Node attribute:* adult depression status.

3. Smirnov, I., & Thurner, S. (2017). "Formation of homophily in academic performance: Students change their friends rather than performance." ***PLOS ONE*, 12(8): e0183473.** *Nodes:* individual high-school and university students (Russian sample, ∼6,200 students). *Edges:* directed "like" ties from a social-networking site, observed at 3-month intervals over up to 3.5 years. *Node attribute:* GPA, the main outcome variable. *Method:* (i) a lagged regression showing a friend's past GPA does not predict a student's future GPA once their own past GPA is controlled for, ruling out direct peer influence; (ii) a comparison of GPA gaps on newly-formed vs. dropped ties, showing selection (not influence) drives the GPA-similarity of friend groups; (iii) a simple agent-based model reproducing the observed homophily dynamics. **This is the dataset and the paper this project builds on and replicates.** Its central limitation, and the gap this project fills, is that its network is used only to compute a first-order, ego-level statistic (the average GPA of a student's direct out-links) — it never uses the network's broader topology, and therefore cannot distinguish "popular" from "well-reached."

The Full Chain of Reasoning

This section documents the actual analytical path taken, including the dead ends, because the corrections along the way are part of what makes the final conclusion defensible.

Step 1 — Replicate the original paper on its own data

Using the same replication data (Harvard Dataverse, doi:10.7910/DVN/SZA9YW), we rebuilt Smirnov & Thurner's two core checks in Python. Both replicated cleanly: a lagged regression of GPA_t on GPA_{t-1} and $\overline{GPA}^{friends}_{t-1}$ (high school only, the one cohort with a real GPA time series) found the friend-GPA lag insignificant (coefficient -0.0055, $p = 0.557$) once own past GPA is controlled for (coefficient on own lag: 0.955, $p < 0.001$) — friends' past performance does not predict a student's future performance. Separately, new ties showed a significantly smaller GPA gap than dropped ties in the pooled university sample (gap difference -0.021, $p < 0.001$) but not significantly so in high school (-0.010, $p = 0.183$) — confirming Prediction 3: university's greater social flexibility produces a stronger, cleaner selection effect.

Step 2 — Add Katz–Bonacich centrality naively; get an encouraging but naive result

Adding Katz–Bonacich centrality (computed per network snapshot, $\alpha = 0.85 \times 1/\lambda_{max}$) to a pooled regression of GPA on average-friend-GPA and cohort fixed effects gave a significant, positive coefficient (standardized $\beta = 0.093$, $p < 0.001$). Read uncritically, this looked like clean support for Prediction 2.

Step 3 — Stress-test it: does this survive a horse race against raw degree?

Katz–Bonacich centrality correlates 0.85–0.89 with simple in-degree in these networks — it could just be a complicated repackaging of "how many people like you." Running the same regression with raw in-degree and out-degree included alongside Katz–Bonacich caused the Katz coefficient to **flip sign and lose most of its significance** (from +0.093, $p < 0.001$ to a small, unstable estimate), while in-degree remained robustly significant and R^2 barely moved. **The naive Step-2 result does not survive this check** — it was mostly popularity in disguise. This is reported as an honest negative finding, not smoothed over.

Step 4 — Decompose Katz–Bonacich by its attenuation parameter

Katz–Bonacich centrality has a tunable decay parameter α: as $\alpha \to 0$ it is numerically indistinguishable from raw in-degree; as α approaches its theoretical ceiling ($1/\lambda_{max}$) it increasingly reflects long, indirect, multi-step reach through the network. Sweeping α from 5% to 95% of that ceiling showed the correlation with in-degree falling from 0.9999 to 0.753, and — critically — the Katz coefficient in a horse race against raw degree moving from a positive, degree-driven artifact at low α ($\beta = +0.082$, $p = 0.012$) through a null zone, to an **independent, robust, negative** effect at high α ($\beta = -0.048$, $p = 0.031$ at $\alpha = 0.80$; $\beta = -0.036$, $p = 0.011$ at $\alpha = 0.95$). Betweenness centrality — a structurally distinct "bridging" measure, less collinear with degree ($r = 0.70$ vs. 0.87 for Katz) — showed the same negative, independent sign in a combined horse race with all four centrality measures at once ($\beta = -0.059$, $p < 0.001$), while eigenvector centrality, like naive Katz, was absorbed entirely by degree.

α (frac. of max)	0.05	0.20	0.40	0.60	0.80	0.95
Corr. with in-degree	0.9999	0.9976	0.9884	0.9657	0.9053	0.7530
Katz coef. (horse race)	+0.082	+0.022	−0.034	−0.051	−0.048	−0.036
p-value	0.012	0.598	0.407	0.115	**0.031**	**0.011**

Table 1: Attenuation-parameter sweep: as α rises, Katz–Bonacich diverges from raw popularity and an independent negative effect emerges.

Step 5 — Lock down one pre-specified headline test

Because Step 4's sweep tested six α levels, reporting "significant at 4 of 6" would overstate the evidence — adjacent α values are highly correlated, so this is closer to one smooth trend crossing a threshold than four independent replications. To avoid this, we treat as **confirmatory** the one specification fixed *before* the sweep was ever run: the $\alpha = 0.85$ baseline established when the panel was first built, tested once, pooled across the full sample, controlling simultaneously for raw in/out-degree, average friend GPA (Smirnov & Thurner's own selection-channel variable), and cohort fixed effects:

$$GPA_{it} = \beta_0 + \beta_1\,\text{Katz}^z_{it} + \beta_2\,\text{InDegree}^z_{it} + \beta_3\,\text{OutDegree}^z_{it} + \beta_4\,\overline{GPA}^{friends}_{it} + \gamma_{\text{cohort}} + \varepsilon_{it}$$

Katz–Bonacich coefficient $\hat{\beta}_1 = -0.0415$ (SE $= 0.019$, $p = 0.033$, clustered by student, $n = 36{,}696$).

Because $\overline{GPA}^{friends}$ — the exact variable that carries Smirnov & Thurner's selection mechanism — is already in this regression, a residual, independently significant Katz effect on top of it is direct evidence that network position carries information their method cannot access, not merely a relabelling of their own finding.

Step 6 — Test Prediction 4 properly, across the whole α range

At the pre-specified $\alpha = 0.85$ alone, only two of five cohorts (high school, $\beta = -0.085$, $p = 0.021$; juniors, $\beta = -0.100$, $p = 0.008$) showed a significant effect; the pooled "university" category was not significant as a whole at that single α. Running the same test across the full α sweep showed a more complete picture:

Cohort	α=0.05	α=0.20	α=0.40	α=0.60	α=0.80	α=0.95
High school	−0.010	−0.075	**−0.129***	**−0.128***	**−0.099***	**−0.071***
Juniors	+0.029	−0.107	**−0.194***	**−0.194***	**−0.156***	**−0.110***
Freshmen	−0.087	−0.198	−0.118	−0.064	−0.021	+0.012
Sophomores	+0.059	+0.112	+0.120	+0.103	+0.078	+0.046
Seniors	**+0.194***	**+0.190***	+0.109	+0.039	−0.003	−0.021

Table 2: Katz coefficient (horse race vs. degree) by cohort across the full α sweep. $^*p < 0.05$.

High school is significantly negative across four of six α levels, the single most consistent result in the dataset — directly reversing an earlier premature reading that high school showed no effect. **Pooled university**, by contrast, is only significant at the extreme high end ($\alpha = 0.95$: $\beta = -0.032$, $p = 0.032$). Breaking university down further by class year shows the pooled result is not a clean "university" story at all: **juniors** replicate the high-school pattern closely; **sophomores** run consistently in the *opposite* direction at every α level (never significant, but never once crossing zero); **freshmen and seniors** are ambiguous, likely underpowered (freshmen has only 2 network snapshots). **Prediction 4, as originally framed around a clean high-school/university divide, is rejected** — the real heterogeneity tracks specific cohorts, not the level distinction.

Step 7 — A structural explanation for the cohort heterogeneity

Ranking the five cohorts by average in-degree (network sparsity) and comparing to each cohort's Katz coefficient at $\alpha = 0.95$ shows a striking pattern: the two cohorts with the most robust negative effect (school: avg. in-degree 3.84; juniors: 3.95) are the two **sparsest** networks in the dataset; the two cohorts running the opposite direction (sophomores: 5.06; freshmen: 5.61) are among the **densest**. Correlation across the five cohort-level points: $r = 0.86$ ($n = 5$ — suggestive, not a basis for statistical inference, but a plausible mechanism: in a sparse network, high indirect reach can only be achieved by genuinely spreading thin across disconnected pockets, whereas in a dense network the same reach may simply reflect being embedded in one large, well-connected cluster, without the same attention-dilution cost).

Step 8 — Benchmark the effect size honestly

Recomputing Smirnov & Thurner's own headline statistic — the Homophily Index, Pearson's r between a student's GPA and their friends' average GPA, using their exact published formula on the same data — gives a pooled mean of $r \approx 0.315$ (by cohort: school 0.304, freshmen 0.250, sophomores 0.298, juniors 0.303, seniors 0.344), consistent with a medium-to-strong, well-established effect. The regression-scale analogue in our own headline model, the coefficient on $\overline{GPA}^{friends}$, is $\mathbf{0.476}$ GPA points per GPA point of friend average — a large, dominant effect. Against that backdrop, our headline Katz effect of $-\mathbf{0.0415}$ standard deviations is genuinely small — on the order of one-tenth the size of the selection channel the original paper already established.

Conclusion

Confirmed: the original paper's core selection-vs-influence logic replicates on this data (Prediction 1), and university's greater social flexibility produces a measurably stronger selection effect than high school's (Prediction 3).

Revised, not confirmed as originally stated: network position does carry independent information beyond friend-group composition (Prediction 2) — but only once "network position" is correctly disentangled from raw popularity via the attenuation parameter; a naive centrality regression does not survive scrutiny. **Rejected:** the effect does not divide cleanly along high school vs. university (Prediction 4) — it is cohort-specific, plausibly tracking network sparsity rather than social-environment stability.

The honest headline finding: being directly liked by many people (popularity) and being reachable through many indirect, loosely-connected paths (extended network reach) are empirically and statistically separable positions in a friendship network, and they point in *opposite* directions for GPA — popularity helps, extended reach mildly hurts, holding popularity fixed. This distinction is structurally invisible to Smirnov & Thurner's method, which only ever measures a student's immediate friends' average GPA and could never have surfaced it. The effect is modest in size — roughly a tenth as large as the friend-selection effect the original paper documents — and is not universal across every sub-sample we tested; both limitations are reported directly rather than smoothed over.

Why this is still worth reporting. The value here is not the size of one coefficient. It is (i) a worked demonstration of separating a network's *structural position* effect from its *compositional* effect — a genuinely hard identification problem in the peer-effects literature — using a tool this paper's own method structurally cannot access; (ii) a concrete, transferable technique (decomposing a centrality measure by its own attenuation parameter to isolate local popularity from extended reach) that could be reapplied to other networks entirely; and (iii) a finding that pushes back on a common, unexamined assumption — that a "bigger network" or "more connections" is straightforwardly good — with a specific, testable counter-example: reach without depth is not obviously an asset, and in the sparsest networks in this data, it is a measurable liability.

Data: Smirnov & Thurner (2017) replication data, Harvard Dataverse, doi:10.7910/DVN/SZA9YW. Full analysis pipeline: `scripts/friendship_gpa/01--04`*. All regression outputs referenced above are saved under* `output/friendship_gpa/`*.*